RESEARCH PROTOTYPE · NOT CLINICALLY VALIDATED

HypoSense Twin

An exposure-aware digital twin that estimates whether a CGM-detected low-glucose episode will begin within the next hour — so staff can check the patient earlier.

Development evidence only · the pre-registered confirmatory evaluation is pending.

Lows are easy to miss between scheduled glucose checks

- Adults with type 2 diabetes on insulin or sulfonylureas can develop low glucose (hypoglycaemia).
- On a ward, capillary glucose is usually checked on a timetable; an episode can start in between.
- A continuous glucose monitor (CGM) shows where glucose *is* — not where it is heading in the next hour.

Goal: warn **15–60 minutes before** a CGM-detected low begins, at a fixed budget of about **one false alert per patient-day**.

2 · INSIGHT

Medicines keep acting after they are given

Insulin and secretagogues act for hours. HypoSense tracks the **recorded doses** and estimates how much may still be **active**, using documented, label-sourced pharmacological profiles.

Recorded
dose and time, as entered in the record

→

Estimated activity
a model-based curve over the next hours — never a measured blood level, never a dosing suggestion

→

Twin feature
"medicine activity now and soon", available to the forecast

Scientific question: does this exposure information improve early warning beyond CGM + context?

One causal pipeline for replay and live data

Data
CGM, doses, meals, context — file replay, CSV/XLSX upload or manual entry

→

Validation
one canonical parser; nothing guessed, nothing silently dropped

→

Patient state
glucose history, data quality, medicine exposure, forecast ledger

→

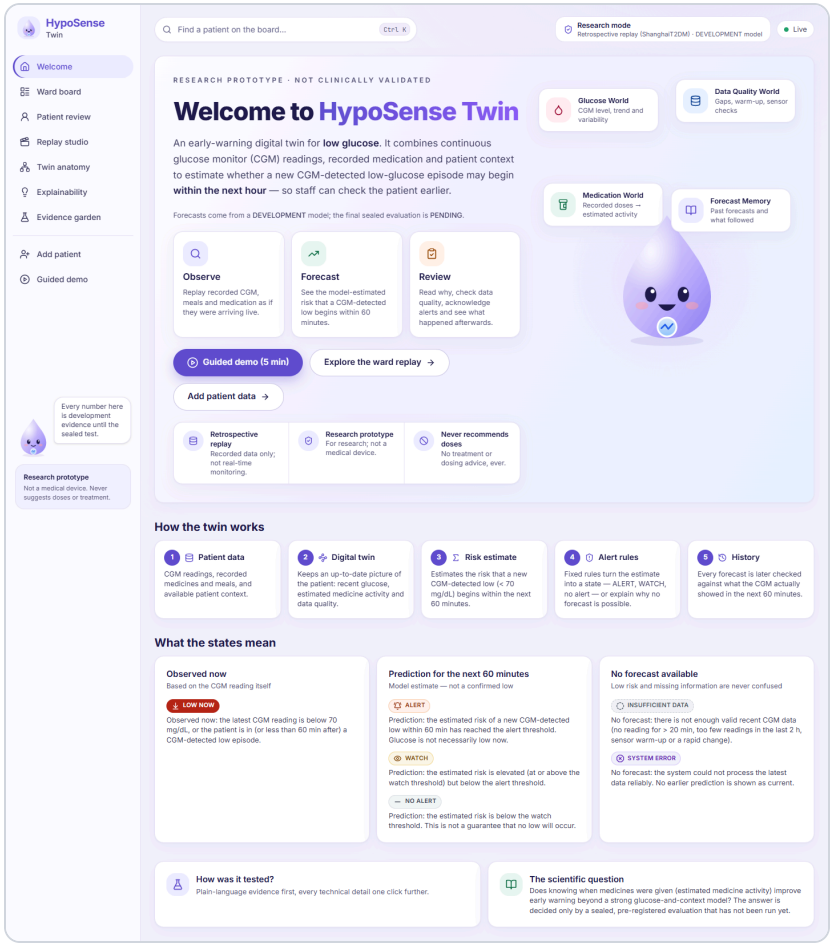
Forecast
100 time-valid features → LightGBM → calibrated risk

→

Fixed rules
ALERT · WATCH · NO ALERT · LOW NOW · or "no forecast"

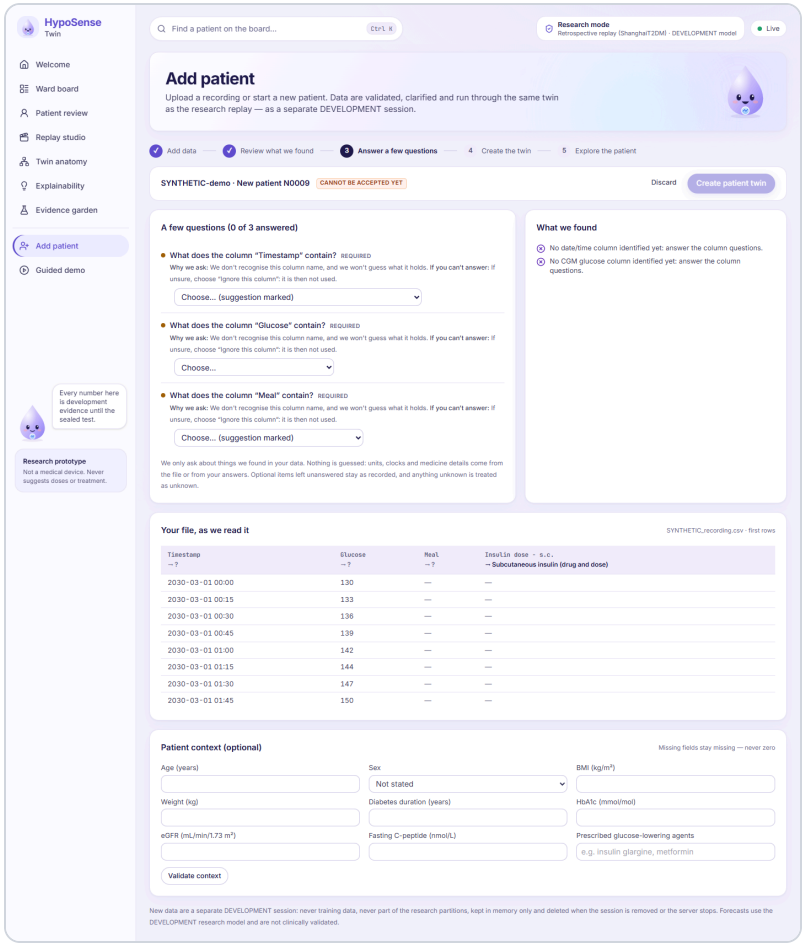
- Features use only information available at the prediction time — verified bitwise at every development time point.
- "Low risk" and "not enough data" are different states and are never confused.
- Every forecast is later checked against what the CGM actually showed.

Understandable in 30 seconds



- **Welcome** — what it is, the four parts of the twin, what each state means
- **Guided demo** — 8 steps on the real replay
- **Ward board** — who needs attention now
- **Patient review** — observed now vs predicted next 60 min; twin anatomy; why; what happened afterwards
- **Add patient** — upload or enter data; clarifying questions
- **Evidence garden** — plain language first, every number on demand

New data: it asks, it never guesses



- Upload CSV/XLSX or start an empty patient.
- Questions only where the data need them — column meaning, glucose unit, clock, unreadable medicine entries — each with *why we ask* and *what if you can't answer*.
- Same parser and twin as the research data (parity-tested).
- Never training data; kept in memory; no accuracy claim for uploaded patients.

Screenshot: SYNTHETIC data.

6 · THE EXPERIMENT

One pre-registered question, a fair ladder, a sealed test set

MODEL	ADDS
M0	glucose trend rule (no learning)
M1	learned CGM features
M2	+ patient context and meals
M3r	+ raw medicine timing
M3	+ estimated medicine exposure

H1: M3 warns more episodes than M2 at 1 false alert per patient-day in treated patients of the sealed partition — supported only if the 95% paired patient-bootstrap interval of the difference is above 0.

Patients split once before modelling: 30 development · 70 sealed (never opened so far).

Development data: no reliable difference yet

SENSITIVITY @ 1 FA/DAY (TREATED, 22 PATIENTS, 56 EPISODES)	ESTIMATE	95% CI
M0 trend rule	62.6 %	49.1 – 83.7
M1 CGM	85.7 %	71.9 – 100
M2 + context	87.5 %	78.1 – 100
M3 + exposure	91.1 %	78.8 – 100

M3 − M2 = **+3.6 points (−4.2 to +8.8)**: the interval includes zero. Tuning happened on these patients, so numbers are optimistic. CONFIRMATORY: PENDING

Source: run T-AGG-dev-s6-20261018T144427Z.

8 · HONESTY

False alerts are shown, not hidden

- In the guided demo, the first real alert (risk 15 %) was **not** followed by a low — the history shows it as such.
- Labels are CGM-detected; CGM and capillary glucose often disagree around 70 mg/dL.
- One public dataset, one centre, internal validation, small numbers.
- Medicine profiles await clinical-pharmacist review.
- Never recommends doses or treatment; never calls a patient "safe".

Built so the result cannot be tuned after the fact

- Sealed evaluation data are refused by the code until a verified protocol freeze exists; the confirmatory run executes once.
- Freeze records hashes of code, configuration, data, partition, analysis plan, locks and model.
- Model rebuilt repeatedly with an identical results hash; replay decision logs are identical run to run.
- 572 Python tests, 42 frontend tests, 3 browser end-to-end tests; 0 accessibility violations (axe, WCAG 2 A/AA).

10 · STATUS AND NEXT STEPS

Engineering complete · science one step from the decisive test

- **Done:** data audit, labels, exposure engine, twin, ladder, policy, replay, API, frontend, new-patient intake, confirmatory runner (rehearsed on development data).
- **Pending (human):** clinical-pharmacist review of medicine profiles; owner approval of the confirmatory operational definitions.
- **Then:** protocol freeze → one confirmatory run on the sealed patients → model selection by the pre-registered rule → report, whatever the result.

Data: ShanghaiT2DM (Zhao et al., Scientific Data 2023; CC BY 4.0).